

Nessus Essentials Scan Report

Objective

Perform a vulnerability assessment of Metasploitable2 using Nessus Essentials.

Tools Used

Tool	Version	Purpose
Nessus Essentials	10.12.2	Automated vulnerability assessment and security analysis
VMware Workstation Pro	Latest	Virtualization platform hosting the virtual machines
Kali Linux	2025.x	Scanner machine used to perform the assessment
Metasploitable2	Ubuntu-based vulnerable VM	Target machine containing intentionally vulnerable services

Target Information

Target Name: Metasploitable2

Target IP Address: 192.168.180.131

Operating System: Ubuntu Linux (Metasploitable2)

Environment:

- Scanner Machine: Kali Linux
- Target Machine: Metasploitable2
- Both virtual machines running inside VMware Workstation Pro.

Scan Configuration

Scanner - Nessus Essentials

Scan Template - Basic Network Scan

Target Host - 192.168.180.131

Scan Type - Unauthenticated Vulnerability Scan

Authentication - No credentials were provided.

Plugin Feed - Latest Nessus Essentials plugin feed updated prior to scanning.

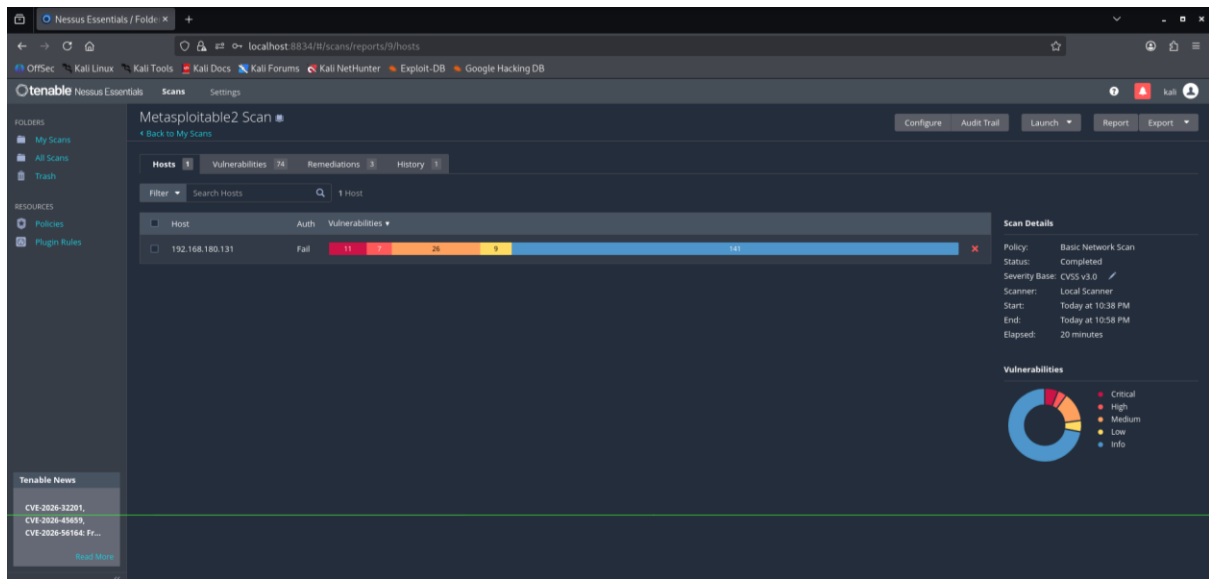
Methodology

1. An initial Nmap scan was performed to discover the target host and identify open ports and running services.
2. Nessus Essentials was configured using the **Basic Network Scan** template.
3. The target IP (192.168.180.131) was scanned for known vulnerabilities.
4. Scan results were analyzed based on severity levels (Critical, High, Medium, Low, and Informational).
5. Appropriate remediation recommendations provided by Nessus were reviewed.

Scan Results

Severity	Count
Critical	11
High	7
Medium	26
Low	9
Informational	141

The scan identified a total of **194 findings**, including **11 Critical** and **7 High** severity vulnerabilities. These results indicate that the target system contains multiple outdated services, insecure configurations, and known backdoors that could allow an attacker to gain unauthorized access or compromise the system.



Scan Summary

Sev	CVSS	VPR	EPSS	Name	Family	Count
CRITICAL	10.0			Canonical Ubuntu Linux SEoL (8.04.x)	General	1
CRITICAL	10.0 *			UnrealIRCd Backdoor Detection	Backdoors	1
CRITICAL	10.0 *			VNC Server 'password' Password	Gain a shell remotely	1
CRITICAL	9.8			SSL Version 2 and 3 Protocol Detection	Service detection	2
CRITICAL	9.8			Bind Shell Backdoor Detection	Backdoors	1
MIXED				Apache Tomcat (Multiple Issues)	Web Servers	4
CRITICAL				SSL (Multiple Issues)	Gain a shell remotely	3
HIGH	7.5			NFS Shares World Readable	RPC	1
HIGH	7.5 *			rlogin Service Detection	Service detection	1
HIGH	7.5 *			rsh Service Detection	Service detection	1
HIGH	7.5			Samba Badlock Vulnerability	General	1
MIXED				SSL (Multiple Issues)	General	28

Scan Details

- Policy: Basic Network Scan
- Status: Completed
- Severity Base: CVSS v3.0
- Scanner: Local Scanner
- Start: July 30 at 10:38 PM
- End: July 30 at 10:58 PM
- Elapsed: 20 minutes

Vulnerabilities

- Critical
- High
- Medium
- Low
- Info

Vulnerabilities List

Top Critical Vulnerabilities

1. Canonical Ubuntu Linux SEoL (8.04.x)

Severity: Critical (CVSS 10.0)

Description

The operating system is Ubuntu 8.04, which has reached End-of-Life (EoL). It no longer receives security updates or patches.

Risk

An attacker can exploit publicly known vulnerabilities that remain unpatched, leading to system compromise.

Recommendation

Upgrade to a currently supported Ubuntu LTS release.

2. UnrealIRCd Backdoor Detection

Severity: Critical (CVSS 10.0)

Description

The installed UnrealIRCd server contains a known malicious backdoor.

Risk

A remote attacker can execute arbitrary commands on the target system.

Recommendation

Remove the compromised version and install a clean, updated release.

3. VNC Server 'password' Password

Severity: Critical (CVSS 10.0)

Description

The VNC server uses the default password "password".

Risk

Attackers can easily gain remote desktop access without sophisticated attacks.

Recommendation

Replace default credentials with a strong unique password and restrict remote access.

4. SSL Version 2 and 3 Protocol Detection

Severity: Critical (CVSS 9.8)

Description

The server supports outdated SSLv2 and SSLv3 protocols.

Risk

These protocols are vulnerable to attacks such as POODLE and other cryptographic weaknesses.

Recommendation

Disable SSLv2 and SSLv3 and enable only TLS 1.2 or TLS 1.3.

5. Bind Shell Backdoor Detection

Severity: Critical (CVSS 9.8)

Description

A bind shell backdoor is listening on the system.

Risk

An attacker can directly connect to the backdoor and execute commands remotely.

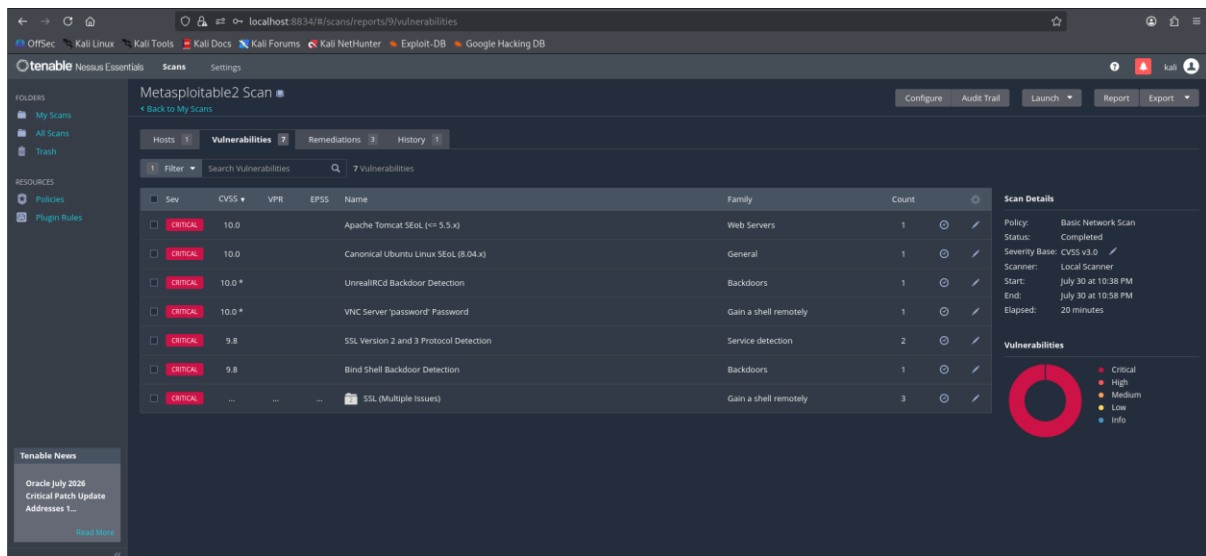
Recommendation

Remove the backdoor, investigate the system for compromise, and secure exposed services.

Overall Analysis

The assessment revealed numerous high-risk vulnerabilities on the Metasploitable2 virtual machine. Most Critical findings were caused by intentionally vulnerable software, default credentials, outdated operating system versions, insecure network protocols, and known backdoors. These issues could allow remote code execution, unauthorized remote access, privilege escalation, and complete system compromise.

The scan also reported **3 remediation groups**, indicating that multiple vulnerabilities can be resolved by applying a small number of corrective actions such as upgrading software, removing insecure services, and disabling obsolete protocols.



Sev	CVSS	VPR	EPSS	Name	Family	Count
CRITICAL	10.0			Apache Tomcat SEOL (CVE-2021-40279)	Web Servers	1
CRITICAL	10.0			Canonical Ubuntu Linux SEOL (B.04.x)	General	1
CRITICAL	10.0 *			UnrealIRCd Backdoor Detection	Backdoors	1
CRITICAL	10.0 *			VNC Server 'password' Password	Gain a shell remotely	1
CRITICAL	9.8			SSL Version 2 and 3 Protocol Detection	Service detection	2
CRITICAL	9.8			Bind Shell Backdoor Detection	Backdoors	1
CRITICAL	...	...	...	SSL (Multiple Issues)	Gain a shell remotely	3

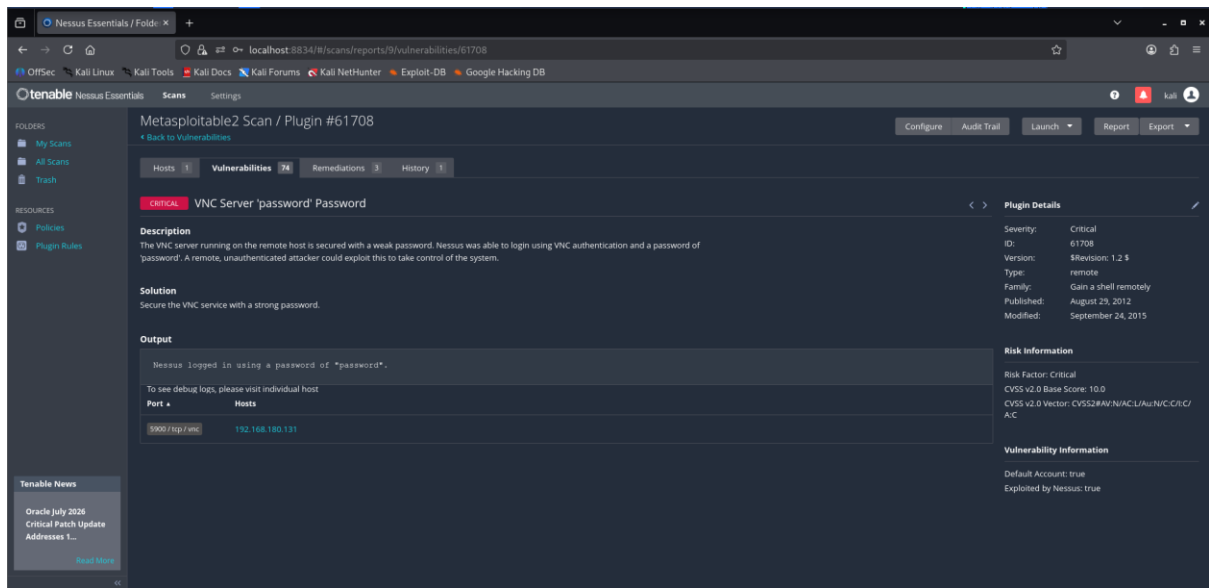
Scan Details

- Policy: Basic Network Scan
- Status: Completed
- Severity Base: CVSS v3.0
- Scanner: Local Scanner
- Start: July 30 at 10:38 PM
- End: July 30 at 10:58 PM
- Elapsed: 20 minutes

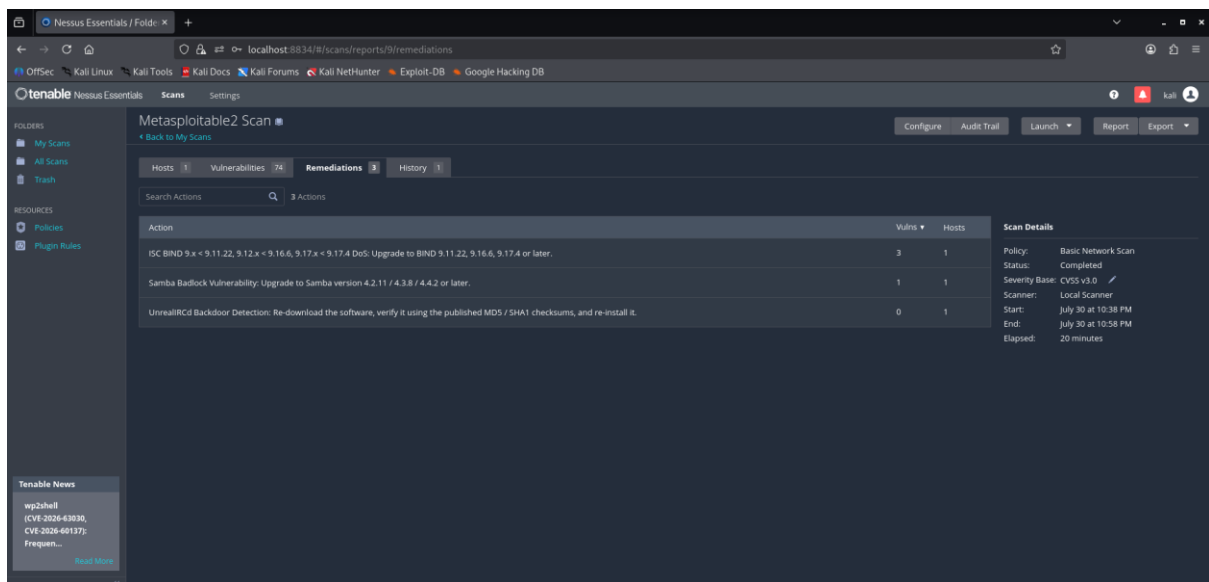
Vulnerabilities

- Critical (Red)
- High (Orange)
- Medium (Yellow)
- Low (Green)
- Info (Blue)

Critical Vulnerabilities



VNC Server 'password' Password



Remediation

Conclusion

The vulnerability assessment successfully identified several critical security weaknesses within the Metasploitable2 environment. Nessus Essentials categorized the findings by severity and provided remediation guidance. The results demonstrate how outdated software, insecure configurations, and default credentials significantly increase an organization's attack surface. Applying the recommended remediations would substantially improve the security posture of the system.